

Priced Certificates for Conditional-CVaR Obedience

Abstract

For a fixed finite-signal policy, we study whether one realised batch of conditional-loss samples can certify exact obedience under upper-tail conditional value-at-risk (CVaR). A simultaneous confidence box and a finite difference-constraint graph give either a supporting state-independent reward, a negative-cycle witness, or abstention. Under a strict cycle margin, the least nonnegative reward cap exceeds the true-utility oracle cap by at most an explicit confidence term. Sampling cost can therefore be charged alongside one reward payment. This is a local, confidence-qualified result; it is not a competitive ratio for a learning-augmented system.

1 Setting and provenance

The supplied Q is a survey of learning-augmented algorithms. It proposes a charge per predictor invocation and gives sufficient conditions for composing component cost guarantees, including conditional guarantees at reachable histories and relations among benchmarks [8]. The supplied P studies a one-type, finite-signal obedience test with state-independent action rewards [7]. Its expected empirical CVaR surrogate can reverse the true obedience decision at every fixed finite batch size. P also proves deterministic cycle and common-reward tests when all conditional utilities lie within a stated error bound. P explicitly leaves one realised-batch certification outside those results. The mechanism paper called Q inside P is distinct from the supplied Q.

We fix a finite signal set S , a finite set B of available actions, and a policy $a : S \rightarrow B$ chosen before sampling. Every available action is prescribed at some signal; alternatively, unprescribed actions are barred. The agent has one type and evaluates an action after observing s by $U(b, s) = -\text{CVaR}_\beta(L_{s,b})$, where $\beta \in (0, 1]$ is upper-tail mass and $L_{s,b} \in [0, H]$. A state-independent reward $r : B \rightarrow \mathbb{R}$ supports the policy when

$$U(a(s), s) + r(a(s)) \geq U(b, s) + r(b) \quad \text{for every } s \in S \text{ and } b \in B.$$

This reduced-form conditional-risk model excludes ex-ante CVaR of a plan, report-stage incentives, and the system objective considered by Q.

The contribution below is the realised-data certificate and its priced, local reward-cap comparison. P's graph theorem supplies the deterministic step; the CVaR concentration step and margin reasoning have close published antecedents discussed in Section 5.

2 A certificate from one realised batch

For each of the $N = |S||B|$ cells (s, b) , assume access to m independent, identically distributed draws from its fixed conditional loss law. Dependence between different cells is allowed. Write $\widehat{U}(b, s)$ for negative upper-tail CVaR of that cell's realised empirical distribution. For $0 < \delta < 1$, set

$$e_m = \frac{H}{\beta} \sqrt{\frac{\log(2N/\delta)}{2m}}, \quad \mathcal{E} = \left\{ \max_{s,b} |\widehat{U}(b, s) - U(b, s)| \leq e_m \right\}.$$

The variational CVaR formula gives $|\text{CVaR}_\beta(F) - \text{CVaR}_\beta(G)| \leq H\|F - G\|_\infty/\beta$ for distributions on $[0, H]$. The Dvoretzky–Kiefer–Wolfowitz inequality and a union bound yield $\Pr(\mathcal{E}) \geq 1 - \delta$. This bounds the error of a realised batch relative to true CVaR; P's expected empirical CVaR is a different quantity.

For distinct $a, b \in B$, define the empirical robust edge and the true edge by

$$w_m(a, b) = \max_{s: a(s)=a} \{\widehat{U}(b, s) - \widehat{U}(a, s) + 2e_m\},$$

$$w^*(a, b) = \max_{s: a(s)=a} \{U(b, s) - U(a, s)\}.$$

The graph has vertex set B and one such edge for each ordered pair of distinct actions. A cycle's weight is the sum of its edge weights.

Theorem 1 (Realised-batch decision). *If the empirical robust graph has no positive cycle, its difference constraints admit a vector r satisfying*

$$r(a) - r(b) \geq w_m(a, b) \quad (a \neq b).$$

The recorded estimates, radius, graph and vector then certify exact true obedience on $\mathcal{E}$. If instead a displayed cycle of K edges uses actions $a_1, \dots, a_K, a_{K+1} = a_1$ and specified signals s_k with $a(s_k) = a_k$, define

$$\widehat{C} = \sum_{k=1}^K [\widehat{U}(a_k, s_k) - \widehat{U}(a_{k+1}, s_k)].$$

If $\widehat{C}$ satisfies $\widehat{C} + 2Ke_m < 0$, it certifies true infeasibility on $\mathcal{E}$. In every other case the procedure abstains.

Proof. For every signal prescribing a , the event $\mathcal{E}$ implies $U(b, s) - U(a, s) \leq w_m(a, b)$. Summing difference constraints around a cycle proves their necessity; P's path-potential construction proves sufficiency when there is no positive cycle. The resulting r satisfies every true obedience inequality on $\mathcal{E}$. Each utility difference in a cycle changes by at most $2e_m$, so the second inequality makes the corresponding true cycle negative. Such a cycle cannot coexist with any supporting reward. The graph step is P's deterministic rectangular-error test, applied to a random box with stated coverage. $\square$

The certificate is sound with probability at least $1 - \delta$ over the specified sample draw. A positive robust cycle alone proves neither true infeasibility nor failure of a particular true reward: it can lead to abstention. A negative-cycle test uses an upper confidence bound, not the robust graph's failed feasibility check.

3 Margin, payment and sample price

Assume every nontrivial simple true cycle in the graph has weight at most $-K\gamma$, where K is its length and $\gamma > 0$. This is equivalent to a positive per-edge margin for the corresponding

utility-cycle test. On $\mathcal{E}$, direct comparison of edge maxima gives

$$w^*(a, b) \leq w_m(a, b) \leq w^*(a, b) + 4e_m.$$

Thus $e_m < \gamma/4$ makes every empirical robust cycle negative, and Theorem 1 produces a reward witness. One sufficient uniform batch size is

$$m > \frac{8H^2}{\beta^2\gamma^2} \log \frac{2N}{\delta}.$$

A true cycle with utility sum at most $-K\gamma$ likewise produces a negative certificate under this radius. These are sufficient margin conditions, not optimal sample-complexity claims.

We now require nonnegative rewards. Let $n = |B|$, let R^* be the smallest possible cap $\max_a r(a)$ for a reward satisfying the true constraints, and let R_m^* be the corresponding cap for the empirical robust constraints. A cap is the maximum payment for one decision, not the expected payment under a specified signal distribution.

Proposition 1 (Local reward-cap comparison). *Under the preceding cycle margin, on $\mathcal{E}$ and when $e_m < \gamma/4$,*

$$R^* \leq R_m^* \leq R^* + 4(n-1)e_m.$$

If one conditional-loss draw costs κ_s in the decision objective's units, sampling plus at most one reward payment is bounded by

$$\kappa_s Nm + R^* + 4(n-1)e_m$$

on the coverage event, apart from any separately charged graph computation.

Proof. For any edge array without a positive cycle, the coordinatewise least nonnegative solution of $r(a) - r(b) \geq w(a, b)$ is the maximum weight of a path starting at a , including the empty path. A maximising path can be simple, so it uses at most $n-1$ edges. The edge comparison above raises any such path weight by at most $4(n-1)e_m$. Taking maximum coordinates of these least solutions gives the stated cap comparison. $\square$

The payment term is necessary. For instance, edge requirements $r(a) - r(b) \geq M$ and $r(b) - r(a) \geq -M - \gamma$ have a negative cycle, yet any nonnegative supporting reward has cap at least M . A declared budget R can be checked by adding $0 \leq r(a) \leq R$ to the robust difference constraints. Failure of that budget check does not show that the true array violates the budget.

For $\kappa_s > 0$, the sufficient upper bound in Proposition 1 has variable part $\kappa_s Nm + Am^{-1/2}$, where

$$A = 4(n-1) \frac{H}{\beta} \sqrt{\frac{\log(2N/\delta)}{2}}.$$

Its continuous unconstrained minimiser is $m_0 = (A/(2\kappa_s N))^{2/3}$. The margin condition additionally requires the strict batch-size inequality above; an integer batch is chosen above that threshold and near the larger of the threshold and m_0 . This only optimises the displayed sufficient bound.

Q charges predictor invocations, whereas the bill here charges draws. Treating one draw as one invocation, or one batch as one invocation, is an extra accounting choice. Neither R^* nor $\kappa_s Nm + R_m^*$ is Q's prediction-free offline system benchmark.

4 Two access boundaries

The sample oracle must reveal losses for deviations at each signal, not just losses for the prescribed action. In a two-signal swapped-action instance, give each prescribed action constant loss $3/2$ in both worlds. Give every unplayed deviation loss 2 in one world and 1 in another. Every on-policy transcript has the same law in both worlds, for any sample size, while the true two-edge utility cycles have weights $+1$ and -1 . Thus on-policy logs alone cannot identify the obedience decision. A generative conditional sampler, covered exploration, or a justified identification model is a separate premise whose cost must be counted.

There is also no uniform finite-sample exact yes-or-no decision near the cycle boundary. Let the safe loss be 1, risky loss be $2X$ with $X \sim \text{Bernoulli}(p)$, and $\beta = 1/2$. In two signals, swap which named action is risky and prescribe the risky action at each signal. For $p < 1/2$, risky CVaR is $4p$, giving true utility-cycle sum $C(p) = 2 - 8p$. At $p_{\pm} = 1/4 \pm \varepsilon$, with $0 < \varepsilon < 1/4$, the signs differ. Every fixed finite binary transcript has positive probability in both worlds. A nonabstaining sequential test with errors at most $\delta < 1/2$ in both worlds and finite expected informative sample count T obeys

$$\mathbb{E}_{p_-} T \text{kl}(p_-, p_+) \geq \text{kl}(1 - \delta, \delta).$$

The per-draw divergence is of order ε^2 , so expected sampling cost grows as the margin shrinks. This obstruction permits abstention and positive-margin promises.

5 Related work and interpretation

Concentration for empirical CVaR is established by Prashanth, Jagannathan and Kolla [2]; a Wasserstein treatment of empirical risk estimation that includes CVaR appears in Prashanth and Bhat [1]. Agrawal, Koolen and Juneja [3] give fixed-confidence best-arm methods and confidence bounds for tail-risk measures. These works publish the statistical main step used here. P [7] supplies the deterministic graph step in the supplied manuscript. Thus the contribution is their narrow combination with a priced reward witness and local cap comparison, rather than a new concentration or graph theorem.

Chen [5] uses a positive risk-value margin to retain strict incentive compatibility under approximation in a different persuasion model. Balcan, Sandholm and Vitercik [4] estimate approximate incentive compatibility from samples for other mechanism classes. The sequential change-of-measure ingredient behind the boundary example is standard in best-arm identification [6]. None of these sources, in the targeted searches recorded in [search.md](#), was found to state the specific conditional-CVaR reward-cap comparison above; that search does not establish publication priority.

6 Limitations and open questions

The local guarantee concerns a fixed policy, bounded conditional losses, sample access for all compared cells, state-independent rewards and one decision. It makes no claim for ex-ante CVaR, report-stage incentives, unobserved counterfactuals, or a general learning-augmented pipeline. The graph construction gives a high-confidence statement, not a pathwise guarantee on failed coverage. A negative-cycle witness and a supporting reward are sound only on the coverage event.

An expected-cost claim needs an abstention action and failure-path cost control. If downstream cost $D \leq B$ on the entire coverage event, including abstention, and $D \leq M$ on every path, then

$$\mathbb{E}[\kappa_s Nm + D] \leq \kappa_s Nm + B + \delta M.$$

For Q’s adaptive composition, both coverage and cost premises must hold conditionally on each reachable pre-sampling history. A common positive offline benchmark and component-to-system cost relations would also be needed for a competitive ratio. None is supplied by this local result.

Open questions are how to price and justify counterfactual access, how to allocate samples around critical graph cycles, and how transfers and a bounded abstention fallback enter a common downstream objective. A first system-level test would specify these elements in the two-signal setting and prove, or refute, a conditional expected-cost comparison with its prediction-free benchmark.

References

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